Video Style Transfer Evaluation - Synthetic Workflow Demonstration

Claim status: Exploratory
Date: 2026-11-08
Claim: Demonstrate a reproducible style-content-motion evaluation workflow without presenting synthetic clips as model benchmark evidence.

Executive verdict

SSIM ↑: Simulated balanced 0.9290 vs Simulated flicker 0.8111 (better; n=3).

LAB palette W1 ↓: Simulated balanced 9.0208 vs Simulated flicker 7.4201 (worse; n=3).

OFFSET 1 WARP RMSE ↓: Simulated balanced 31.70 vs Simulated flicker 47.77 (better; n=3).

OFFSET 1 FLOW EPE ↓: Simulated balanced 0.2438 vs Simulated flicker 0.5426 (better; n=3).

Runtime ↓: Simulated balanced 18.80 vs Simulated flicker 15.00 (worse; n=3).

Blinded VLM review: Simulated balanced has the highest mean overall visual quality (4.00/5 across 1 reviewer(s)); this is perceptual evidence, not benchmark ground truth.

Most frequent visual failure tag: detail loss (3 occurrence(s)).

The diagnostic style metric measures palette alignment only; texture and brushwork claims require perceptual evidence.

Evidence gates

Gate	Value	Status
Paired valid cases	3	Pass
Methods	3	Pass
Blinded VLM review	Present	Pass
Paper-grade perceptual metrics	Not detected	Not established
Claim status	Exploratory	Caveat

LAB palette distance and Farneback flow are deterministic diagnostic proxies. They are not RAFT, LPIPS, CLIP, FID, or ArtFID.

Metric definitions and motivation

SSIM

Availability: Computed in this run

```
SSIM(x,y) = ((2*mu_x*mu_y + C1)*(2*sigma_xy + C2)) / ((mu_x^2 + mu_y^2 + C1)*(sigma_x^2 + sigma_y^2 + C2))
```

Motivation: SSIM compares local luminance, contrast, and structure instead of raw pixel error. It is useful for detecting geometric and structural changes, but strong stylization legitimately changes contrast and texture.

Interpretation and reporting caveat: Interpret it as a content diagnostic, not a perceptual style score.

PSNR

Availability: Computed in this run

```
PSNR = 10 * log10(MAX^2 / MSE)
```

Motivation: PSNR converts mean squared error into a logarithmic scale and gives a quick, deterministic fidelity diagnostic. It is sensitive to global color shifts, compression, and resolution differences.

Interpretation and reporting caveat: Do not treat a small PSNR change as a proportional change in perceived quality.

LPIPS

Availability: Not provided in this run; definition retained for paper-grade extension

```
LPIPS(x,y) = sum_l (1/(H_l*W_l)) * sum_hw ||w_l * (phi_l(x)_hw - phi_l(y)_hw)||^2
```

Motivation: LPIPS compares normalized deep features and aligns better with perceptual similarity than pixel metrics. It is preferred as preservation evidence when a fixed LPIPS implementation is available.

Interpretation and reporting caveat: The network, preprocessing, and weights must be identical across every compared method.

CLIP-F

Availability: Not provided in this run; definition retained for paper-grade extension

```
CLIP-F(R,I) = 100 * cos(phi_I(R), phi_I(I))
```

Motivation: CLIP-F measures whether the stylized and original frames retain the same image-level semantics. It complements structure metrics when appearance changes substantially.

Interpretation and reporting caveat: Image-level embeddings can miss local identity, boundary, and occlusion errors.

CLIP-I

Availability: Not provided in this run; definition retained for paper-grade extension

```
CLIP-I(R,S) = 100 * cos(phi_I(R), phi_I(S))
```

Motivation: CLIP-I provides a content-robust global signal for whether the generated frame and style reference occupy a similar semantic visual space. It is one component of style evidence, not a texture-transfer proof.

Interpretation and reporting caveat: Report it comparatively and supplement it with texture-specific features or blinded review.

FID

Availability: Not provided in this run; definition retained for paper-grade extension

```
FID = ||mu_s - mu_g||_2^2 + Tr(S_s + S_g - 2*(S_s*S_g)^(1/2))
```

Motivation: FID compares Inception feature distributions, so it can capture style-set mismatch beyond single-image color statistics. It requires a meaningful distribution of reference and generated frames.

Interpretation and reporting caveat: A single style image is not a valid FID style distribution. Record image count, resolution, and preprocessing.

ArtFID

Availability: Not provided in this run; definition retained for paper-grade extension

```
ArtFID = (1 + mean_i LPIPS(I_i,R_i)) * (1 + FID(R,S))
```

Motivation: ArtFID combines perceptual content preservation and style-distribution distance in one number. The product makes a method visible when it improves one factor while degrading the other.

Interpretation and reporting caveat: Do not use it unless its LPIPS and FID components follow the cited, fixed implementation.

LAB palette W1

Availability: Computed in this run

```
W1_c = sum_k |CDF_frame,c(k) - CDF_style,c(k)| / 255; PaletteW1 = 100 * mean_c W1_c
```

Motivation: This inexpensive diagnostic summarizes marginal LAB color alignment. It is useful for detecting global recoloring and palette mismatch before expensive perceptual metrics.

Interpretation and reporting caveat: It does not measure brushwork, texture attachment, composition, or style semantics.

Source-flow warping RMSE

Availability: Computed in this run

```
WarpRMSE_d = sqrt(sum_x M_t(x)*||R_t(x) - W(R_(t+d))(x)||_2^2 / sum_x M_t(x))
```

Motivation: Source optical flow supplies the expected motion. Warping a later output frame back with that flow isolates drift and flicker from legitimate scene motion; occluded pixels are masked.

Interpretation and reporting caveat: The result inherits the flow estimator's errors; Farneback is a diagnostic proxy, not RAFT.

Flow endpoint error

Availability: Computed in this run

```
EPE_d = (1/|Omega|) * sum_{x in Omega} ||F_R(x) - F_I(x)||_2
```

Motivation: EPE measures whether stylization changes the estimated displacement between frames. It is sensitive to motion alteration and can reveal propagation failures that still images miss.

Interpretation and reporting caveat: Texture changes can create spurious flow; use the same estimator and occlusion handling for every method.

Flow-field correlation

Availability: Computed in this run

```
rho_d = corr(flatten(F_R), flatten(F_I))
```

Motivation: Correlation measures agreement in motion direction independently of the flow magnitude emphasized by EPE. Together they separate directional corruption from displacement scaling.

Interpretation and reporting caveat: It is a global summary and may hide localized motion errors.

Temporal LPIPS (tLP)

Availability: Not provided in this run; definition retained for paper-grade extension

```
tLP = (1/(T-d)) * sum_t LPIPS(R_t, W(R_(t+d)))
```

Motivation: tLP replaces pixel error in warping analysis with a perceptual deep-feature error, making texture and structure instability easier to detect than raw L2 differences.

Interpretation and reporting caveat: Use a fixed LPIPS network and report the temporal offset.

CLIP temporal consistency

Availability: Not provided in this run; definition retained for paper-grade extension

```
CLIP-CONS = 100 * cos(phi(R_t+1)-phi(R_t), phi(I_t+1)-phi(I_t))
```

Motivation: CLIP-CONS asks whether semantic change between adjacent stylized frames follows the semantic change in the source. It complements pixel warping when content is stylized heavily.

Interpretation and reporting caveat: It measures semantic change agreement, not spatial alignment or flicker.

Detrended luminance flicker

Availability: Computed in this run

```
F = sqrt((1/T) * sum_t (l_t - (a*t+b))^2), where l_t is frame mean luminance
```

Motivation: Removing a linear luminance trend distinguishes intentional exposure or scene changes from abrupt global flicker. The measure is cheap and directly targets one common VST artifact.

Interpretation and reporting caveat: It ignores local flicker and texture sticking, which require motion-aware or visual review.

Blinded VLM score

Availability: Computed in this run

$$S_{d,m} = \frac{1}{R} \cdot \sum_r \frac{1}{|C_{valid}|} \cdot \sum_c s_{r,c,m,d},\ s \in \{1,2,3,4,5\}$$

Motivation: The rubric captures style adherence, content preservation, temporal stability, artifact control, and overall quality when automated metrics cannot separate texture and motion failures. Per-case blinded labels reduce method-identity bias.

Interpretation and reporting caveat: VLM judgments are perceptual evidence, not ground truth; record model version and use multiple independent reviewers for paper-level claims.

Runtime and peak memory

Availability: Computed in this run

$$Runtime_m = \text{mean}_c\, t_{c,m};\quad PeakMemory_m = \max_c peak_{c,m}$$

Motivation: Quality comparisons are incomplete without deployment cost. Mean runtime captures expected latency, while maximum peak memory captures the worst observed resource requirement.

Interpretation and reporting caveat: Measure on identical hardware, precision, resolution, and batch settings; report both mean and peak.

Quantitative results

Style

Metric	Simulated balanced	Simulated flicker	Simulated overlay
LAB palette W1 (lower)	9.0208 ± 0.5445	**7.4201 ± 0.5347**	<u>8.0416 ± 0.9411</u>
LAB chroma palette W1 (lower)	<u>3.1765 ± 1.3380</u>	**1.8824 ± 1.0178**	4.6879 ± 0.4238

Best and second-best mean values are shown in bold and underlined, respectively. Valid clips: balanced: 3; flicker: 3; overlay: 3.

Content

Metric	Simulated balanced	Simulated flicker	Simulated overlay
SSIM (higher)	**0.9290 ± 0.0165**	<u>0.8111 ± 0.0326</u>	0.7153 ± 0.0149
PSNR (higher)	**22.58 ± 0.9018**	<u>17.54 ± 0.3337</u>	14.90 ± 0.3931

Best and second-best mean values are shown in bold and underlined, respectively. Valid clips: balanced: 3; flicker: 3; overlay: 3.

Temporal and motion

Metric	Simulated balanced	Simulated flicker	Simulated overlay
OFFSET 1 WARP RMSE (lower)	**31.70 ± 6.7038**	<u>47.77 ± 10.70</u>	63.87 ± 6.8744
OFFSET 5 WARP RMSE (lower)	**33.20 ± 5.2688**	<u>46.18 ± 10.55</u>	63.70 ± 7.5960
OFFSET 1 FLOW EPE (lower)	<u>0.2438 ± 0.0558</u>	0.5426 ± 0.3239	**0.1874 ± 0.0547**
OFFSET 1 FLOW CORRELATION (higher)	**0.4268 ± 0.2381**	0.2267 ± 0.0923	<u>0.2662 ± 0.1339</u>
Detrended luminance flicker (lower)	<u>0.1035 ± 0.0619</u>	12.43 ± 0.4689	**0.0822 ± 0.0217**

Best and second-best mean values are shown in bold and underlined, respectively. Valid clips: balanced: 3; flicker: 3; overlay: 3.

Cost

Metric	Simulated balanced	Simulated flicker	Simulated overlay
Runtime (lower)	18.80 ± 0.4000	**15.00 ± 0.4000**	16.50 ± 0.4000
Peak GPU memory (lower)	12.70 ± 0.0000	**9.8000 ± 0.0000**	10.40 ± 0.0000

Best and second-best mean values are shown in bold and underlined, respectively. Valid clips: balanced: 3; flicker: 3; overlay: 3.

Paired comparison against the primary baseline

Metric	d (positive favors primary)	dz	95% CI	n
SSIM	0.1180	7.1536	—	3
LAB palette W1	-1.6007	-15.05	—	3
OFFSET 1 WARP RMSE	16.08	3.9236	—	3
OFFSET 1 FLOW EPE	0.2988	1.0159	—	3
Runtime	-3.8000	n/a	—	3

Blinded visual review

Method	Style adherence	Content preservation	Temporal stability	Artifact control	Overall visual quality
Simulated balanced	4.00/5	4.67/5	3.67/5	3.33/5	4.00/5
Simulated flicker	4.67/5	3.67/5	2.00/5	2.00/5	2.33/5
Simulated overlay	3.33/5	3.00/5	3.00/5	2.00/5	2.00/5

Failure tags: detail loss (3), texture sticking (3), flicker (3), edge ghosting (2), temporal jump (2), texture drift (1), geometry melting (1)

Reviewer records: 1. Scored case/method records: 9. VLM scores are perceptual evidence, not ground truth.

Analysis and discussion

Style fidelity

The available indicators collectively favor Simulated flicker. Reported diagnostics: LAB palette W1 9.0208 ± 0.5445; LAB chroma palette W1 3.1765 ± 1.3380. The blinded style-adherence score is 4.00/5.

Content preservation

The available indicators collectively favor Simulated balanced. Reported diagnostics: SSIM 0.9290 ± 0.0165; PSNR 22.58 ± 0.9018.

Motion and temporal stability

The available indicators collectively favor Simulated balanced. Reported diagnostics: OFFSET 1 WARP RMSE 31.70 ± 6.7038; OFFSET 5 WARP RMSE 33.20 ± 5.2688; OFFSET 1 FLOW EPE 0.2438 ± 0.0558; Detrended luminance flicker 0.1035 ± 0.0619.

Perceptual review and failure modes

The blinded review gives the highest overall visual quality to Simulated balanced (4.00/5 across 1 reviewer(s)). This reflects preference on the reviewed sample rather than ground-truth quality. The leading failure modes are detail loss (3), texture sticking (3), flicker (3).

Computational cost

The available indicators collectively favor Simulated flicker. The primary method uses runtime 18.80 ± 0.4000 s and peak memory 12.70 ± 0.0000 GB.

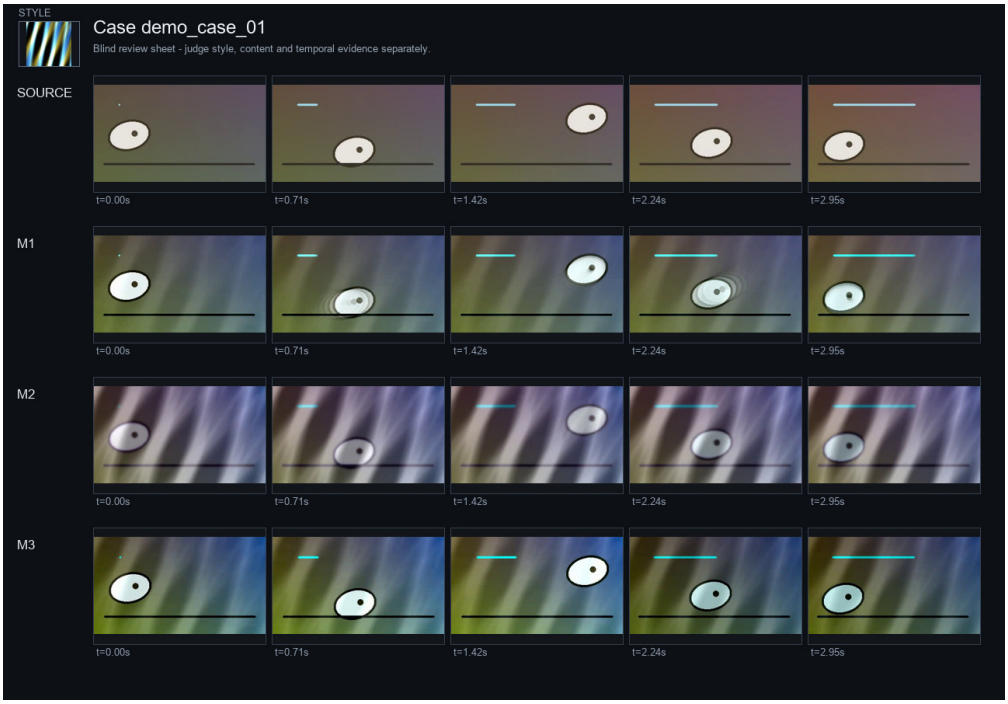
Synthesis

The results show a multidimensional trade-off rather than universal dominance; the current evidence does not support a claim that one method is uniformly superior.

Per-case outcomes

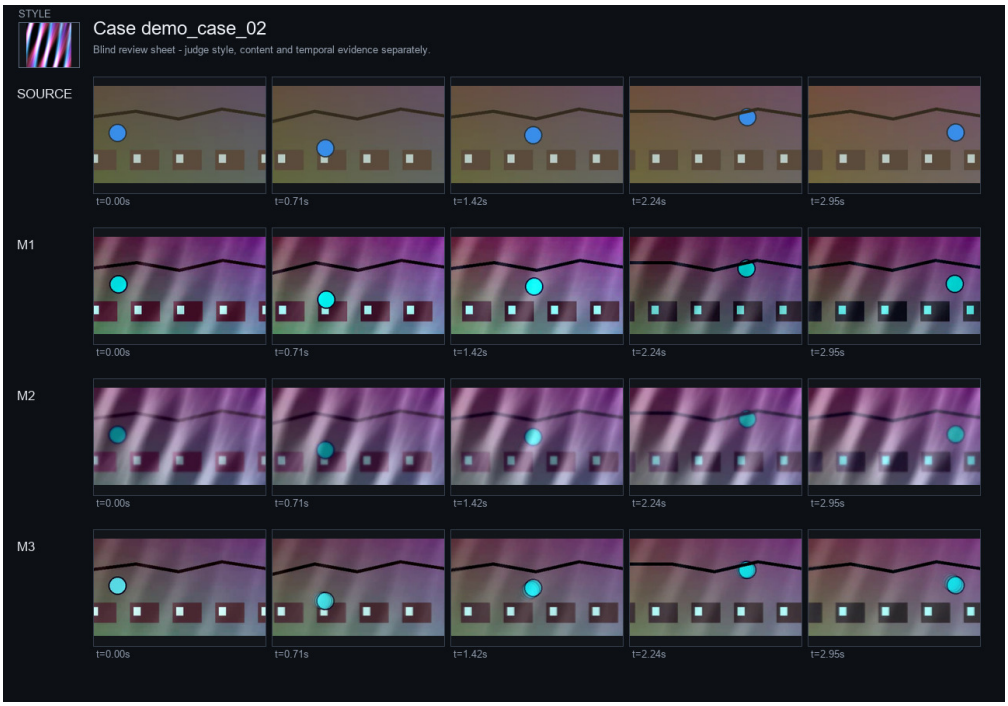
demo_case_01

Method	SSIM (higher)	Palette W1 (lower)	Warp RMSE d1 (lower)	Flow EPE d1 (lower)
Simulated balanced	0.9395	8.8136	28.61	0.2963
Simulated flicker	0.8270	7.3344	44.06	0.5494
Simulated overlay	0.7122	8.1708	65.43	0.1389



demo_case_02

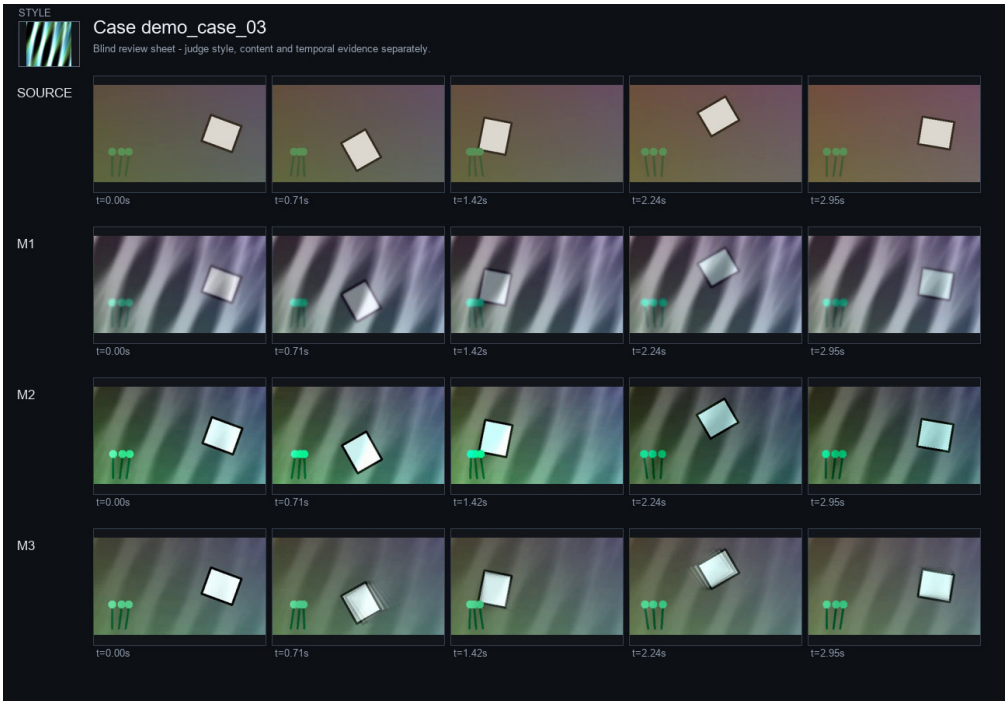
Method	SSIM (higher)	Palette W1 (lower)	Warp RMSE d1 (lower)	Flow EPE d1 (lower)
Simulated balanced	0.9101	8.6103	39.39	0.2500
Simulated flicker	0.7736	6.9334	59.84	0.8630
Simulated overlay	0.7315	7.0425	69.83	0.2467



demo_case_03

Method	SSIM (higher)	Palette W1 (lower)	Warp RMSE d1 (lower)	Flow EPE d1 (lower)
Simulated balanced	0.9376	9.6385	27.09	0.1852

Method	SSIM (higher)	Palette W1 (lower)	Warp RMSE d1 (lower)	Flow EPE d1 (lower)
Simulated flicker	0.8327	7.9925	39.42	0.2154
Simulated overlay	0.7022	8.9114	56.35	0.1766



Threats to validity

- Content and style distributions may not represent the intended use.
- Low-level metrics are sensitive to colorization and compression.
- Farneback flow is not RAFT and can fail on large motion or occlusions.
- A single style image does not define a valid FID style distribution.
- VLM review can have position, content, and prompt bias.
- External baseline numbers copied from papers may not use the same settings.
- Fewer than five clips do not support a stable comparative confidence interval.

Conclusion and recommendation

This report uses 3 synthetic cases to validate the evaluation pipeline and does not provide evidence of real model performance. The tables, perceptual analysis, and conclusions preserve multidimensional evidence and should not be interpreted as a benchmark ranking of EchoStyle or the proposed method.

Recommendation

Use these results to validate the evaluation pipeline and choose the next experiment. Do not treat the synthetic run as model evidence.